BUSINESS INTELLIGENCE

BASIC OF DATA WAREHOUSE

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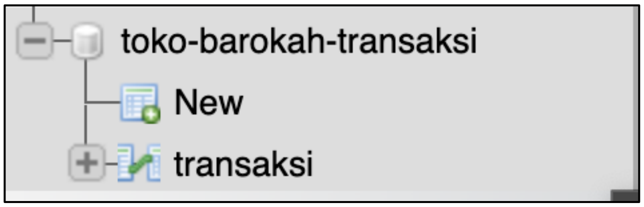
PRACTICUM PART 1 – Setting Up Large Database Examples



PRACTICUM PART 2 – Preparing an Application Sample

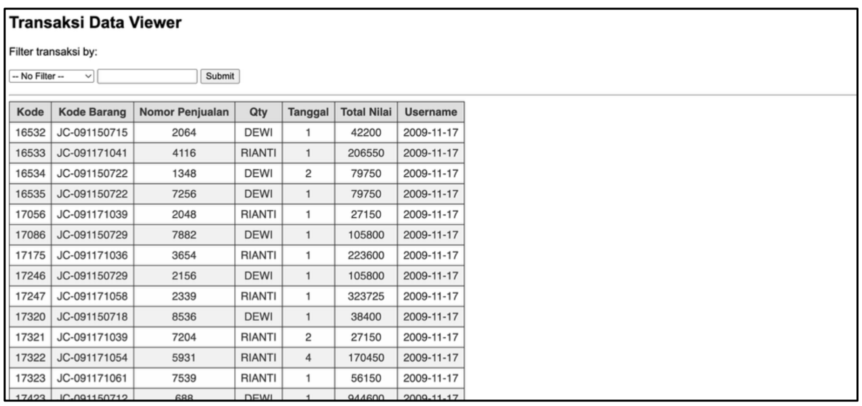
Transaksi Data Viewer

Filter transaksi by:

-- No Filter -- Submit

Kode	Kode Barang	Nomor Penjualan	Qty	Tanggal	Total Nilai	Username
No records to show. Please click Submit button or try another filter.						

PRACTICUM PART 3 - Observing Application Performance with Large Amounts of Data

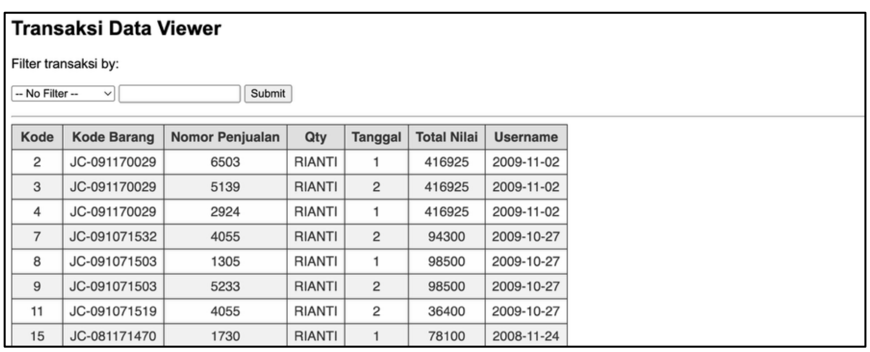


Transaksi Data Viewer

Filter transaksi by:

-- No Filter -- Submit

Kode	Kode Barang	Nomor Penjualan	Qty	Tanggal	Total Nilai	Username
16532	JC-091150715	2064	DEWI	1	42200	2009-11-17
16533	JC-091171041	4116	RIANTI	1	206550	2009-11-17
16534	JC-091150722	1348	DEWI	2	79750	2009-11-17
16535	JC-091150722	7256	DEWI	1	79750	2009-11-17
17056	JC-091171039	2048	RIANTI	1	27150	2009-11-17
17086	JC-091150729	7882	DEWI	1	105800	2009-11-17
17175	JC-091171036	3654	RIANTI	1	223600	2009-11-17
17246	JC-091150729	2156	DEWI	1	105800	2009-11-17
17247	JC-091171058	2339	RIANTI	1	323725	2009-11-17
17320	JC-091150718	8536	DEWI	1	38400	2009-11-17
17321	JC-091171039	7204	RIANTI	2	27150	2009-11-17
17322	JC-091171054	5931	RIANTI	4	170450	2009-11-17
17323	JC-091171061	7539	RIANTI	1	56150	2009-11-17
17423	JC-091150712	688	DEWI	1	944600	2009-11-17

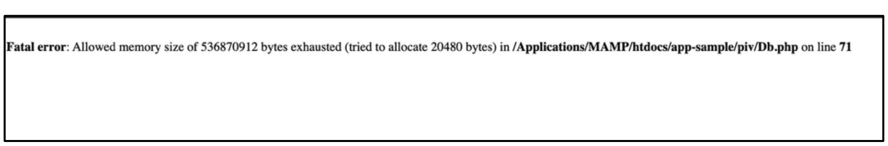


Transaksi Data Viewer

Filter transaksi by:

-- No Filter -- Submit

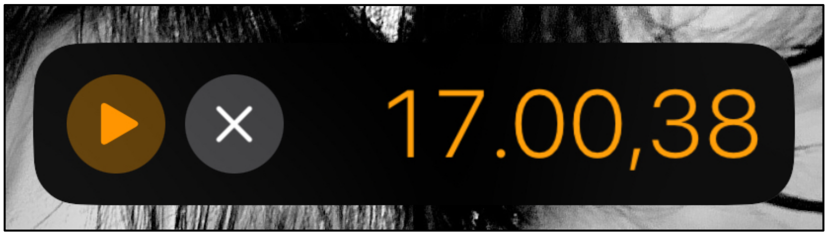
Kode	Kode Barang	Nomor Penjualan	Qty	Tanggal	Total Nilai	Username
2	JC-091170029	6503	RIANTI	1	416925	2009-11-02
3	JC-091170029	5139	RIANTI	2	416925	2009-11-02
4	JC-091170029	2924	RIANTI	1	416925	2009-11-02
7	JC-091071532	4055	RIANTI	2	94300	2009-10-27
8	JC-091071503	1305	RIANTI	1	98500	2009-10-27
9	JC-091071503	5233	RIANTI	2	98500	2009-10-27
11	JC-091071519	4055	RIANTI	2	36400	2009-10-27
15	JC-081171470	1730	RIANTI	1	78100	2008-11-24



Fatal error: Allowed memory size of 536870912 bytes exhausted (tried to allocate 20480 bytes) in **/Applications/MAMP/htdocs/app-sample/piv/Db.php** on line **71**

TRAINING TASK

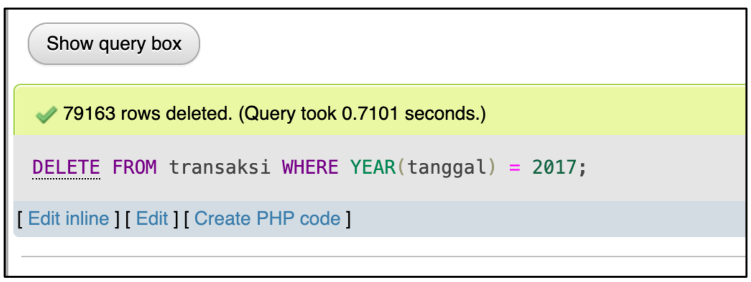
Answer Number 1



As seen in the image above, the server took 17 minutes to respond to the client's request before the loading indicator stopped, out of a total of 443 thousand transaction data.

Answer Number 2

Here is the query for Delete all data from 2017.



DELETE FROM transaksi WHERE YEAR(tanggal) = 2017;

Answer Number 3

The time decreased by 5 minutes from 17 minutes to 12 minutes after the data decreased by nearly 80 thousand.

Answer Number 4

Fatal error: Allowed memory size of 536870912 bytes exhausted (tried to allocate 4096 bytes) in **/Applications/MAMP/htdocs/app-sample/piv/Db.php** on line **70**

In the first error, PHP failed to allocate an additional 4096 bytes (4KB) when the data had been reduced by about 80 thousand, while in the second error (Answer Number 1), PHP failed to allocate 20480 bytes (20KB) when the data was still full at almost 443 thousand. This means that it is not the raw data size that determines when an error occurs, but rather how the data is handled in memory (for example, when looping, fetching queries, or forming arrays/objects). So even though the amount of data is smaller, the memory requirements for certain processes is different.

Answer Number 5

In conclusion, when we have a large amount of data, we cannot display all of it at once, as this can cause the server to crash, preventing client requests from being processed.

Therefore, instead of displaying all the data directly, we can use several optimization techniques to prevent the server from crashing, such as pagination, data caching, etc. Additionally, to anticipate time limits when the server is executing, we can add time limits according to our needs, and we can also add memory allocation on the server to handle requests. However, basically, the time limits and memory allocation provided by each programming language are more than sufficient.